

Executive Summary:

Host Vocabulary Predicts Where New Concepts Take Root

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Headline

Host-vocabulary share predicts newcomer uptake. When an emerging scientific concept enters a new subfield, the share of its initial co-occurrence partners that belong to the host’s native vocabulary predicts whether newcomer authors in that subfield will subsequently adopt the concept. The held-out co-primary estimate is $IRR/SD = 1.19 [1.06, 1.33]$ ($p = 0.004$); MeSH replication gives $IRR/SD = 1.23 [1.12, 1.36]$ (Holm $p < 0.001$). Verdict: **REPLICATED** for biomedicine-to-biomedicine entries. The effect is on how much newcomer uptake follows, not on a binary establishment threshold, and its out-of-sample predictive gain on the held-out fold is not established. For emergence (RQ1), the pre-registered held-out test returns R1_DEAD: structural precursors of sustained uptake are volume/churn correlates.

Goal

The study asks two questions about how emerging scientific concepts are born and spread. RQ1: which temporal and structural patterns in an evolving scientific knowledge graph characterise the emergence of concepts? RQ2: how do emerging concepts diffuse across disciplinary communities, and which network patterns distinguish locally concentrated concepts from broadly integrated ones?

The investigation uses an OpenAlex-based corpus of 426 semantically grounded concepts (462,812 works), constructs yearly co-occurrence networks ($\sim 27,000$ nodes, $\sim 84,000$ edges), and tests structural predictors of emergence and cross-disciplinary integration. A held-out concept fold (100 main concepts after grounding) and an independent MeSH biomedical population (191 concepts) provide confirmation and replication.

What Was Tried

Data foundation (Iteration 1). Built a concept pool of 184 main emerging and 22 reference concepts (later hydrated to 366 main and 60 reference) across physics, physical sciences, and computer science. Assembled a whole-science background sample (259,716 works), a MeSH confirmation population (191 concepts), and a semantic grounding pipeline.

Viability and precursor event study (Iteration 2). Tested citation-layer viability states (source/sink decomposition) and ran a precursor event study for emergence (closure, accretion, participation).

Mechanism tests (Iteration 3). Decomposed the closure effect (turnover vs brokerage), tested breadth prediction, ran host-entry grafting experiments, and built a diffusion typology.

Confirmation (Iteration 4). Opened held-out folds for grafting and closure. Ran discriminating tests (single-paper artefact, vocabulary decomposition, topic-score absorption). Replicated grafting on MeSH. Tested adopter-level mechanism.

Post-confirmation decomposition (Iteration 5). Decomposed the grafting effect into extensive and intensive margins. Tested host-specificity and field boundaries. Full numerical audit. Positioning dossier.

Key Results

Finding	Estimate	95% CI	Status
Grafting: held-out	IRR/SD 1.19	[1.06, 1.33]	Confirmed
Grafting: MeSH	IRR/SD 1.23	[1.12, 1.36]	Replicated
Grafting: IVW pooled	IRR/SD 1.26	[1.17, 1.36]	$I^2 = 0$
Extensive margin (IVW)	+6.43 pp/SD	[4.76, 8.11]	Exploratory
Intensive margin (IVW)	IRR/SD 1.183	[1.104, 1.267]	Exploratory
Host-specificity (IVW)	Retention 0.969	[0.81, 1.10]	Host-specific (expl.)
No field boundary	Wald $p = 0.638$	—	Not detected
RQ1 general closure (panel)	−0.391	[−0.855, 0.072]	R1_DEAD
Persistent-nbr closure (panel)	−1.073	[−1.858, −0.289]	Confirmed (qualified)
Adopter enrichment	OR 3.09	[2.32, 4.28]	Screen only
Expansion → diffusion	25/26 concepts	—	Conditional

Co-transfer (entering with origin companions rather than host-native ones) is null in every specification tested.

The adopter-level mechanism (screen fold only) shows that authors who adopt a concept in a new host subfield have higher odds of prior corpus exposure to the concept’s entry partners (OR 3.09 [2.32, 4.28]; prevalence 0.79 vs 0.61), with enrichment strongest for origin-vocabulary (FOREIGN) partners (OR 2.88). This is consistent with absorptive capacity or with topical proximity.

What Failed

1. **Source-sink viability:** Gate A failed at edge level (within-host traced share 17.3%, threshold 50%). The citation-lineage decomposition could not be executed.
2. **General closure:** The primary closure measure did not survive held-out confirmation (R1_DEAD: Holm $p = 0.147$). Only persistent-neighbour closure (top-20 stable associates) survived.
3. **Brokerage interpretation:** Burt constraint is *higher*, not lower, before emergence—opposite to the brokerage prediction.
4. **Breadth prediction:** Openness does not predict where concepts travel (all Holm $p = 0.405$).
5. **Closure–anchoring link:** Null (partial $\rho = 0.025$ screen, 0.053 held-out; both CIs include zero).
6. **Multi-channel typology:** The 3-channel DTW typology is stable only at $k = 2$ (localised vs broad) and adds nothing over an entropy-only typology.

7. **Community roles:** Leiden community roles do not predict host-subfield entry.
8. **Grafting primary specification:** Inconclusive on held-out (IRR/SD 0.98 [0.70, 1.37]; 30 clusters).
9. **Binary establishment:** EST_bin null on held-out (+2.54 pp, $p = 0.16$).

Open Questions

1. **Population generality.** The pool draws overwhelmingly from physics, physical sciences, and computer science. Generalisation to social sciences and humanities is untested.
2. **Single-paper entries.** 87% of co-primary screen events are single-paper entries; on the held-out fold, single-paper entries drive the effect more strongly (interaction $p = 0.004$). The multi-team pooled effect is positive but less precise.
3. **Causal interpretation.** The intensive margin conditions on $Y \geq 1$ (descriptive). The adopter-level vocabulary gradient (FOREIGN > NATIVE) runs opposite to the entry-level grafting finding, suggesting different channels at different scales.
4. **Absorptive capacity vs topical proximity.** The adopter enrichment is consistent with either; the design cannot separate them.
5. **MeSH coverage.** MeSH replication speaks to biomedicine-to-biomedicine entries only (34% of entries removed by coverage rule); the generality controls absorb 12.4% of the effect there (retention 0.88, CI [0.64, 1.04]).
6. **Audit gaps.** 292 numbers in artifact files not transcribed into the report; seeded-perturbation recall 0.60; 4 of 11 Table 18 self-test claims not done.